Mara North Workflows

User Output Guide

About this guide

This document provides a plain-language reference for each workflow: what it does, what inputs it requires, and every file it produces. Each section covers one workflow and ends with a complete listing of output files and the columns contained in any tabular outputs.

1 · TAHMO Weather Report

Workflow: tahmo_weather_report

What this workflow does

The **TAHMO Weather Report** workflow connects to an EarthRanger (ER) server, pulls raw sensor observations recorded by weather stations in the *ER2ER – From GMMF* subject group, and transforms them into a clean daily summary. Seven meteorological variables are extracted from each observation's JSON detail field, aggregated to a per-station daily resolution, and then visualized as interactive line charts. All charts are additionally exported as static PNG images for use in reports.

Required inputs

- **EarthRanger connection** – credentials to the ER server that holds GMMF weather observations.
- **Time range** – start and end datetime defining the reporting period.

How it works (step summary)

#	Step	Detail
1	Fetch observations	Retrieves all subject-group observations from the 'ER2ER – From GMMF' group within the chosen time range.

2	Extract fields	Parses seven numeric fields from the JSON observation-details column: precipitation, surface air temperature, wind speed, wind gusts, soil temperature, relative humidity, and atmospheric pressure.
3	Rename & index	Renames the subject-name column to 'weather_station' and attaches a temporal index aligned to the grouper settings.
4	Daily aggregation	Groups by weather_station + date and computes: sum of precipitation; mean of temperature, wind speed, soil temperature, humidity, and pressure; max of wind gusts.
5	Persist table	Saves the aggregated data as a CSV file.
6	Draw charts	Generates one interactive HTML line chart per meteorological variable, colored by weather station.
7	Convert to PNG	Renders each HTML chart to a 1 280 × 720 px PNG at 2× device scale for high-DPI outputs.

Output files

Output File	Type	Description
`weather_summary_table.csv`	CSV	Daily aggregated weather data for all stations (see column reference below).
`precipitation_readings_over_time.html/.png`	HTML/PNG	Line chart — total daily precipitation (mm) per weather station.
`temperature_readings_over_time.html/.png`	HTML/PNG	Line chart — mean daily surface air temperature (°C) per station.
`wind_speed_readings_over_time.html/.png`	HTML/PNG	Line chart — mean daily wind speed per station.
`wind_gusts_readings_over_time.html/.png`	HTML/PNG	Line chart — maximum daily wind gusts per station.
`soil_temperature_readings_over_time.html/.png`	HTML/PNG	Line chart — mean daily soil temperature (°C) per station.
`relative_humidity_readings_over_time.html/.png`	HTML/PNG	Line chart — mean daily relative humidity per station.
`atmospheric_pressure_readings_over_time.html/.png`	HTML/PNG	Line chart — mean daily atmospheric pressure per station.

Column reference — weather_summary_table.csv

Each row represents one weather station on one calendar date. All numeric values are rounded to two decimal places.

Column	Description
`weather_station`	Name of the EarthRanger subject (sensor / weather station) that recorded the observation.
`date`	Calendar date of the aggregated readings (YYYY-MM-DD).
`precipitation`	Total (sum) precipitation recorded across all observations for that station on that day, in millimetres.
`temperature`	Mean surface air temperature across all observations for that station on that day, in degrees Celsius.
`wind_speed`	Mean wind speed across all observations for that station on that day.
`wind_gusts`	Maximum wind gust value recorded across all observations for that station on that day.
`soil_temperature`	Mean soil temperature across all observations for that station on that day, in degrees Celsius.
`relative_humidity`	Mean relative humidity across all observations for that station on that day, expressed as a percentage.
`atmospheric_pressure`	Mean atmospheric pressure across all observations for that station on that day, in hPa.

Notes

- If no observations are returned for the selected time range the workflow skips all downstream steps gracefully and no output files are created.
- HTML charts are interactive (hover, zoom, pan) and can be opened directly in any modern web browser. PNG files are static exports suitable for documents and presentations.
- Chart resolution is 1 280 × 720 px at 2× scale factor, giving an effective resolution of 2 560 × 1 440px.

2 · Patrol Effort Report

Workflow: patrol_effort_report

What this workflow does

The **Patrol Effort Report** workflow produces a comprehensive picture of ranger patrol activity within MNC conservancy areas. It retrieves all EarthRanger events and patrol observations for the selected time range, processes GPS tracks into trajectories split by transport mode (foot, vehicle, motorbike), calculates effort metrics per ranger and patrol type, generates interactive and static maps showing

patrol routes and coverage density, and computes what percentage of each conservancy area was visited during the period.

Required inputs

- **EarthRanger connection** – credentials to the ER server holding patrol and event data.
- **Time range** – start and end datetime defining the reporting period.
- **MNC conservancy boundaries** – automatically downloaded (GeoPackage).
- **MNC parcels layer** – automatically downloaded (GeoPackage).

How it works (step summary)

#	Step	Detail
1	Load geospatial data	Downloads MNC conservancy boundaries and parcel layers splits the conservancy into grazing zones and builds styled deck.gl layers with colour-coded zone fills and text labels.
2	Retrieve & filter events	Fetches all ER events for the time range. Excludes wildlife distance-count and airstrip events, then calculates daily event totals and a breakdown by event type.
3	Process patrol info	Filters patrol_info events; normalises nested JSON details to extract purpose, participants, authorised-by, and transport type. Summarises count of patrols by purpose.
4	Build patrol tracks	Explodes patrol IDs and participant lists, fetches patrol details and GPS observations from ER, merges with patrol info, and converts observations to relocations. Filters invalid coordinate pairs.
5	Split by transport mode	Separates relocations into foot, vehicle, motorbike, and unspecified streams. Each stream is converted to trajectory segments using mode-appropriate speed and distance filters.
6	Effort metrics per mode	Aggregates trajectory segments by patrol_type_value: unique patrol count, total distance (m → km), total duration (s → hrs), and mean speed. Adds a summary Total row to each table.
7	Per-mode patrol maps	Applies a tab20 colour map by patrol type, creates deck.gl path layers, combines with conservancy boundary and parcel layers, renders interactive HTML maps, and exports each to PNG.
8	Combined trajectories	Merges foot + vehicle + motorbike trajectories into one dataset. Summarises by participant: patrol count, total distance, and total duration. Saves as GeoParquet.
9	Coverage grid & occupancy	Creates a **1 km** patrol coverage grid; counts unique patrol visits per cell; classifies into 5 equal-interval density bins with a RdYlGn colour ramp. Computes the percentage of each conservancy region covered by patrol activity.

10	Persist all outputs	Saves all CSVs, GeoParquet files, HTML maps, and PNGs to the results directory.

Output files

Output File	Type	Description
`total_events_recorded_by_date.csv`	CSV	Daily count of all recorded events (excludes wildlife counts and airstrip events). Includes a Total row.
`total_events_recorded_by_type.csv`	CSV	Daily count of events broken down by event type.
`patrol_purpose_summary.csv`	CSV	Count of patrols grouped by stated purpose. Includes a Total row.
`foot_patrol_efforts.csv`	CSV	Effort metrics for foot patrols grouped by patrol type. Includes a Total row.
`vehicle_patrol_efforts.csv`	CSV	Effort metrics for vehicle patrols grouped by patrol type. Includes a Total row.
`motorbike_patrol_efforts.csv`	CSV	Effort metrics for motorbike patrols grouped by patrol type. Includes a Total row.
`overall_patrol_efforts.csv`	CSV	Per-ranger summary of patrol count, distance, and duration across all transport modes. Includes a Total row.
`patrol_coverage.csv`	CSV	Percentage of each conservancy region covered by patrol activity during the period.
`patrol_relocations.geoparquet`	GeoParquet	All patrol GPS observations as point features, enriched with patrol metadata and participant information.
`patrol_trajectories.geoparquet`	GeoParquet	Combined trajectory segments from all transport modes with renamed patrol metadata columns.
`total_events_recorded.html/.png`	HTML/PNG	Line chart — total number of events recorded per day.
`foot_patrols_map.html/.png`	HTML/PNG	Interactive map of foot patrol routes, colour-coded by patrol type, overlaid on conservancy boundaries.
`vehicle_patrols_map.html/.png`	HTML/PNG	Interactive map of vehicle patrol routes. See note on PNG conversion below.
`motorbike_patrols_map.html/.png`	HTML/PNG	Interactive map of motorbike patrol routes, colour-coded by patrol type.

patrol_coverage_map.html/.png	HTML/PNG	Grid map showing patrol visit density across the conservancy (RdYlGn colour scale, 5 density bins).

Column reference — effort CSVs (foot / vehicle / motorbike)

The three transport-mode effort tables share the same schema. Each row represents one patrol type. A final row labelled **Total** sums numeric columns.

Column	Description
patrol_type_value	Human-readable patrol type label (e.g. 'General Patrol', 'Anti-Poaching').
no_of_patrols	Count of unique patrol IDs of this type.
distance_km	Total distance covered by all trajectory segments for this patrol type, in kilometres.
duration_hrs	Total time spent on patrol across all segments for this patrol type, in hours.
average_speed	Mean speed across all trajectory segments for this patrol type, in km/h.

Column reference — overall_patrol_efforts.csv

Each row represents one ranger (participant). A final row labelled **Total** sums numeric columns.

Column	Description
participants	Ranger name. Rows with no participant recorded appear as 'Unspecified'.
no_of_patrols	Count of unique patrol IDs this ranger participated in.
distance_km	Total distance covered across all patrol segments this ranger was part of, in kilometres.
duration_hrs	Total patrol duration across all segments this ranger was part of, in hours.

Column reference — patrol_coverage.csv

Each row represents one conservancy region. Values are rounded to two decimal places.

Column	Description
name	Name of the conservancy or sub-region.
occupancy_percentage	Percentage of the region's area that contained at least one patrol visit during the reporting period.

Column reference — patrol_relocations.geoparquet

Point-geometry file. Each row is one GPS fix from a patrol observation.

Column	Description
`geometry`	Point geometry (longitude, latitude).
`fixtime`	Timestamp of the GPS fix.
`patrol_id`	Unique identifier of the parent patrol.
`patrol_title`	Display title of the patrol.
`patrol_serial_number`	Sequential serial number assigned to the patrol.
`patrol_start_time`	Date/time the patrol started.
`patrol_end_time`	Date/time the patrol ended.
`patrol_type`	Internal patrol type identifier.
`patrol_type__value`	Human-readable patrol type label.
`patrol_status`	Current status of the patrol (e.g. done, cancelled).
`patrol_subject`	Name of the tracked subject (ranger device).
`participants`	Ranger(s) on the patrol.
`purpose`	Stated purpose of the patrol.
`transport_type`	Mode of transport: foot, vehicle, motorbike, or unspecified.
`junk_status`	Flag indicating whether the fix was considered a junk/erroneous location.
`groupby_col`	Temporal grouping column added by the temporal-index step.

Column reference — patrol_trajectories.geoparquet

Line-geometry file. Each row is one trajectory segment between two consecutive GPS fixes.

Column	Description
`geometry`	LineString geometry of the trajectory segment.
`segment_start`	Timestamp at the start of the segment.
`segment_end`	Timestamp at the end of the segment.
`dist_meters`	Length of the segment in metres.
`timespan_seconds`	Duration of the segment in seconds.
`speed_kmhr`	Speed along the segment in km/h.

`patrol_id`	Unique identifier of the parent patrol.
`patrol_title`	Display title of the patrol.
`patrol_serial_number`	Sequential serial number of the patrol.
`patrol_start_time`	Date/time the patrol started.
`patrol_end_time`	Date/time the patrol ended.
`patrol_type_id`	Internal patrol type identifier.
`patrol_type_value`	Human-readable patrol type label.
`patrol_status`	Current status of the patrol.
`patrol_subject_name`	Name of the tracked subject (ranger device).
`subject_id`	Unique identifier of the tracked subject.
`participants`	Ranger(s) on the patrol.

Notes

- All map steps skip gracefully if their upstream data is empty, so a missing transport mode will not block the rest of the workflow.
- HTML maps are fully interactive (zoom, pan, hover tooltips, camera capture widget) and can be opened in any modern web browser.
- **Vehicle map PNG conversion may time out** — the vehicle patrol layer can be very large and playwright may not finish rendering within the allowed window. If the PNG is missing, open *vehicle_patrols_map.html* in a browser and use the built-in camera widget to export a screenshot manually.
- Map PNGs are rendered at 2x device scale for high-DPI output.
- The 1 km coverage grid classifies visit counts into 5 equal-interval density bins using ared-yellow-green colour ramp (RdYlGn_r): red = least visited, green = most visited.

3 · Logistics Report

Workflow: logistics_report

What this workflow does

The **Logistics Report** workflow processes EarthRanger operational events into three clean tabular summaries covering the main logistical activities recorded at the conservancy: hot-air balloon landings, airstrip passenger movements, and airstrip maintenance work. All outputs are CSV tables — this workflow does not produce maps.

Required inputs

- **EarthRanger connection** – credentials to the ER server holding operational event data.
- **Time range** – start and end datetime defining the reporting period.

How it works (step summary)

#	Step	Detail
1	Fetch events	Retrieves all ER events for the time range; extracts a date column and adds a temporal index.
2	Balloon landings	Filters balloon_landing events; extracts passenger count, balloon company, and lodge from JSON details; cleans bracket notation; fills missing lodges as 'other'; groups by date / company / lodge and sums passengers.
3	Airstrip operations	Filters airstrip_operations events; extracts airline, attendant, camp/lodge, flight number, passenger count, and arrival-or-departure flag; groups by camp and direction, then pivots so each camp is one row with separate Arrival and Departure passenger columns.
4	Airstrip maintenance	Filters airstrip_maintenance events; extracts the maintenance activity type; retains only date and activity columns.
5	Persist tables	Saves each of the three summaries as a CSV file.

Output files

Output File	Type	Description
`balloon_landing_by_date.csv`	CSV	Daily balloon landing summary grouped by company and lodge (see column reference below).
`airstrip_arrivals_and_departure.csv`	CSV	Pivoted airstrip passenger table: one row per camp/lodge, with separate columns for arrivals and departures (see column reference below).

`airstrip_maintenance_table.csv`	CSV	Log of airstrip maintenance activities by date (see column reference below).

Column reference — balloon_landing_by_date.csv

Column	Description
`date`	Calendar date of the balloon landing (YYYY-MM-DD).
`balloon_company`	Name of the balloon operating company.
`lodge`	Lodge or camp where passengers are staying. Blank values are filled as 'other'.
`no_of_passengers`	Total number of passengers recorded for this company/lodge combination on this date.

Column reference — airstrip_arrivals_and_departure.csv

This table is pivoted: rows are camps/lodges and the column names come from the *arrival_or_departure* values recorded in ER (typically **Arrival** and **Departure**). Blank camp values are filled as 'other'.

Column	Description
`camp_lodge`	Name of the camp or lodge .Blank values are filled as 'other'.
`Arrival`	Total number of arriving passengers for this camp during the reporting period.
`Departure`	Total number of departing passengers for this camp during the reporting period.

Column reference — airstrip_maintenance_table.csv

Column	Description
`date`	Calendar date on which the maintenance activity was recorded.
`activity`	Type of maintenance activity performed (sentence case, e.g. Mowing, Grading).

4 · Wildlife Monitoring Report

wildlife_monitoring_report

What this workflow does

The **Wildlife Monitoring Report** workflow produces a comprehensive view of wildlife activity across the MNC conservancy. It processes two categories of events: wildlife incidents (snares, fires, injuries, carcasses, veterinary treatments, illegal grazing) and species-specific sighting records for elephant, buffalo, rhino, lion, leopard, cheetah, giraffe, and hartebeest. For each species the workflow generates

a count summary table and an interactive map; elephant and buffalo additionally produce herd-size distribution maps and bar charts. All maps are exported as PNG images.

Required inputs

- **EarthRanger connection** – credentials to the ER server holding wildlife event data.
- **Time range** – start and end datetime defining the reporting period.
- **MNC conservancy boundaries** – automatically downloaded (GeoPackage).
- **MNC parcels layer** – automatically downloaded (GeoPackage).

How it works (step summary)

#	Step	Detail
1	Load geospatial layers	Downloads and loads MNC conservancy boundaries and parcel layers. Builds styled deck.gl layers for zones, boundaries, parcels, and text labels.
2	Fetch & index events	Retrieves all ER events for the time range and attaches a date column with a temporal index.
3	Wildlife incidents	Filters snare, fire, wildlife carcass/injury/treatment, and illegal grazing events. Normalises JSON details; renames species and cause columns. Saves full records as both CSV and GeoPackage. Generates an incidents summary table and a daily count table.
4	Incidents map	Removes outlier and null geometries; applies a tab20 colour map by incident type; renders an interactive map; exports to PNG.
5	Elephant sightings	Filters elephant_sighting_rep events; extracts herd size, composition, counts by sex/age. Produces a sighting count table, a herd-composition summary, a herd-type map, a herd-size scatter map (dot radius proportional to size), and a herd-size bar chart.
6	Buffalo sightings	Filters buffalo_sighting_rep events; extracts herd type and size. Produces a count table, a herd-type map, a herd-size scatter map, and a herd-size bar chart.
7	Rhino sightings	Filters rhino_sighting_rep events. Produces a count table and a sightings map.
8	Lion sightings	Filters lion_sighting_rep events; extracts pride name, sex/age counts, group size, and individual present. Produces a count table, a unique-prides summary, and a sightings map coloured by pride.
9	Leopard sightings	Filters leopardsightingrep events; extracts individual ID, sex/age counts, group size, and behaviour. Produces a count table, an individual-sightings summary, and a sightings map coloured by individual.

10	Cheetah sightings	Filters cheetah_sighting_rep events; extracts individual ID, sex/age counts, group size, and behaviour. Produces a count table, an individual-sightings summary, and a sightings map coloured by individual.
11	Giraffe sightings	Filters giraffe_sighting events; extracts adult/calf/juvenile counts and total count. Produces a count table and a sightings map.
12	Hartebeest sightings	Filters hartebeest_sighting events. Produces a sightings map (no separate CSV summary).
13	Convert all maps to PNG	Renders all 11 HTML maps to PNG at 2× device scale.

Output files

Output File	Type	Description
`wildlife_events_recorded.csv`	CSV	Full record of all wildlife incidents with normalised detail columns.
`wildlife_events_recorded.gpkg`	GeoPackage	Spatial version of wildlife incident records (point geometry).
`wildlife_incidents_summary_table.csv`	CSV	Incident counts grouped by type (Fire, Snare, Carcass, Injury, Treatment, Illegal grazing).
`wildlife_incidents_recorded_by_date.csv`	CSV	Daily count of wildlife incidents. Includes a Total row; columns: date, no_of_events.
`wildlife_incidents_map.html/.png`	HTML/PNG	Scatter map of incident locations, colour-coded by incident type.
`elephant_events_recorded.csv`	CSV	Elephant sighting count by event type with a Total row.
`elephant_herd_size_distribution_table.csv`	CSV	Total herd-size distribution by herd composition type with a Total row.
`elephant_herd_size_bar_chart.html`	HTML	Bar chart of elephant sighting frequency by herd-size bin. HTML only — not converted to PNG.

Output File	Type	Description
`elephant_sightings_events.html/.png`	HTML/PNG	Map of elephant sightings coloured by herd composition type.
`elephant_herd_types_map.html/.png`	HTML/PNG	Map of elephant sightings with dot size proportional to herd size (Blues colour ramp).
`buffalo_events_recorded.csv`	CSV	Buffalo sighting count by event type with a Total row.

`buffalo_herd_size_bar_chart.html`	HTML	Bar chart of buffalo sighting frequency by herd-size bin. HTML only — not converted to PNG.
`buffalo_herd_map.html/.png`	HTML/PNG	Map of buffalo sightings coloured by herd type (Bachelor, Mixed, Female/calf, Unspecified).
`buffalo_herd_types_map.html/.png`	HTML/PNG	Map of buffalo sightings with dot size proportional to herd size (Blues colour ramp).
`rhino_events_recorded.csv`	CSV	Rhino sighting count by event type with a Total row.
`rhino_sighting_map.html/.png`	HTML/PNG	Map of rhino sighting locations.
`lion_events_recorded.csv`	CSV	Lion sighting count by event type with a Total row.
`unique_lion_prides.csv`	CSV	Count of sightings per named lion pride; columns: lion_pride, no_of_events.
`lion_sightings_map.html/.png`	HTML/PNG	Map of lion sightings coloured by pride name.
`leopard_events_recorded.csv`	CSV	Leopard sighting count by event type with a Total row.
`individual_leopard_summary.csv`	CSV	Count of sightings per individual leopard; columns: individual_present, no_of_events.
`leopard_sightings_map.html/.png`	HTML/PNG	Map of leopard sightings coloured by individual ID.
`cheetah_events_recorded.csv`	CSV	Cheetah sighting count by event type with a Total row.
`individual_cheetah_summary.csv`	CSV	Count of sightings per individual cheetah; columns: individual_present, no_of_events.
`cheetah_sightings_map.html/.png`	HTML/PNG	Map of cheetah sightings coloured by individual ID.
`giraffe_events_recorded.csv`	CSV	Giraffe sighting count by event type with a Total row.
`giraffe_sighting_map.html/.png`	HTML/PNG	Map of giraffe sighting locations.
`hartebeest_sighting_map.html/.png`	HTML/PNG	Map of hartebeest sighting locations. No CSV summary is produced for hartebeest.

Column reference — wildlife_events_recorded.csv / .gpkg

Contains all six incident types in one table. Columns from the JSON detail field are only populated for the relevant event type; they are blank for other rows.

Column	Description
`id`	Unique EarthRanger event identifier.
`time`	Full timestamp of the event.
`date`	Calendar date extracted from the event timestamp.
`event_type`	Event type code (e.g. snare_rep, fire_rep, wildlife_carcass_rep).
`event_category`	Broader category the event belongs to.
`reported_by`	Name of the ranger or user who reported the event.
`geometry`	Point geometry of the event location.
`wildlife_carcass_species`	Species of the carcass (wildlife_carcass_rep events only).
`wildlife_carcass_suspected_cause`	Suspected cause of death (wildlife_carcass_rep events only).
`wildlife_carcass_visible_injury`	Whether visible injuries were present (wildlife_carcass_rep events only).
`wildlife_carcass_comments`	Free-text comments on the carcass (wildlife_carcass_rep events only).
`wildlife_treatment_species`	Species receiving veterinary treatment (wildlife_treatment_rep events only).
`wildlife_treatment_comments`	Free-text treatment notes (wildlife_treatment_rep events only).
`wildlife_treatment_vet_attending`	Name of attending vet (wildlife_treatment_rep events only).
`wildlife_treatment_vet_prognosis`	Vet's prognosis (wildlife_treatment_rep events only).

Column reference — elephant sighting tables

Column	Description
`elephant_sight_herd_composition`	Herd type: Bachelor, Female/calf, Mixed, or Unspecified.

elephant_sight_herd_size	Total number of elephants in the sighted group.
elephant_sight_male	Count of adult males in the group.
elephant_sight_female	Count of adult females in the group.
elephant_sight_sub_adult	Count of sub-adults in the group.
elephant_sight_under_a_year	Count of calves under one year old.
elephant_sight_comments	Free-text observer comments.

Column reference — buffalo sighting tables

Column	Description
buffalo_herd	Herd type: Bachelor, Mixed, Female/calf, or Unspecified.
buffalo_herd_size	Total number of buffalo in the sighted group.

Column reference — lion sighting tables

Column	Description
lion_pride	Pride name (e.g. Acacia, Enkuyanai, Lemek hills, Offbeat, Oldikidiki, River).
individual_present	Name or ID of any individually known lion(s) present.
lion_group_size	Total number of lions in the sighted group.
lion_sight_male	Count of adult males.
lion_sight_female	Count of adult females.
lion_sight_young	Count of cubs/subadults.
lion_behavior	Observed behaviour.

Column reference — leopard & cheetah sighting tables

Column	Description
individual_present	Name or ID of the individual animal. Sentence-cased; blanks filled as 'unknown' (leopard) or 'other' (cheetah).

leopard_group_size / cheetah_group_size	Total number of animals in the sighted group.
leopard_sight_male / cheetah_sight_male	Count of adult males.
leopard_sight_female / cheetah_sight_female	Count of adult females.
leopard_sight_young / cheetah_sight_young	Count of cubs/young.
leopard_behavior / cheetah_behavior	Observed behaviour.

Column reference — giraffe sighting table

Column	Description
total_count	Total number of giraffes in the sighting.
number_of_adults	Count of adult giraffes.
number_of_calves	Count of calves.
number_of_juveniles	Count of juveniles.

Notes

- Each species task block skips gracefully if no matching events are found — other species are unaffected.
- All maps use the ESRI World Hill shade base layer and are auto-zoomed to the MNC grazing zones extent.
- Herd-size scatter maps use dot radius proportional to herd size (Blues colour ramp by size bin) so larger groups appear as larger circles.

Section 5: Livestock Monitoring

Workflow: livestock_monitoring

Overview

The Livestock Monitoring workflow processes cattle and boma-related field reports to produce a picture of livestock distribution and predation pressure across the Mara North Conservancy. It handles three distinct ER event streams: mobile boma location reports (boma positions over time), cattle-count patrols (herd tallies by grazing zone), and livestock predation events (species lost, suspected predator, and location). Each stream produces summary CSV tables and, where spatial data is available, interactive HTML maps exported to PNG.

Required inputs

- EarthRanger credentials and URL (set in workflow parameters).
- Date range: **since** and **until** timestamps.
- MNC conservancy boundaries

How it works (step summary)

#	Step	Detail
1	Load geospatial layers	Downloads and loads MNC conservancy boundaries and parcel layers. Builds styled deck.gl layers for zones, boundaries, parcels, and text labels.
2	Fetch all events	Retrieves all ER events for the time range and attaches a date column with a temporal index for fast sub-filtering.
3	Mobile boma daily count	Filters mobile_boma_rep events. Aggregates by date to produce a daily boma count. Appends a Total row. Saves as total_boma_count_by_date.csv.
4	Cattle-count zone summary	Filters cattle_count events. Cross-tabulates by date and grazing zone (zone_1, zone_2_3, zone_4) to show cattle distribution. Saves as mobile_boma_summary_table.csv.
5	Boma movement map	Renders mobile boma location points on the MNC base layers; exports as interactive HTML and PNG.
6	Livestock predation — full records	Filters livestock_predation_rep events. Normalises JSON detail fields; renames columns. Saves full records as livestock_predation_events.csv and livestock_predation_events.gpkg.
7	Livestock predation — summary	Aggregates predation records by date, suspected predator, livestock species, and number affected. Saves as livestock_predation_summary_table.csv.

#	Step	Detail
8	Predation daily count	Counts predation events per day. Appends a Total row. Saves as livestock_events_recorded_by_date.csv.
9	Livestock predation map	Renders predation event points on the MNC base layers. Uses fixed map coordinates (not global auto-zoom). Exports as HTML and PNG.

Output files

Output File	Type	Description
`total_boma_count_by_date.csv`	CSV	Daily count of mobile boma locations reported. Includes a Total row.
`mobile_boma_summary_table.csv`	CSV	Cattle counts by date and grazing zone (zone_1, zone_2_3, zone_4) with a total_count column.
`boma_movement_map.html/.png`	HTML/PNG	Map of mobile boma movement/location points on MNC base layers.
`livestock_predation_events.csv`	CSV	Full record of all livestock predation events with normalised detail columns.
`livestock_predation_events.gpkg`	GeoPackage	Spatial version of livestock predation records (point geometry).
`livestock_predation_summary_table.csv`	CSV	Predation summary grouped by date, suspected predator, livestock species, and number affected.
`livestock_events_recorded_by_date.csv`	CSV	Daily count of livestock predation events. Includes a Total row.
`livestock_predation_events.html/.png`	HTML/PNG	Map of livestock predation event locations. Rendered at fixed MNC coordinates (not global auto-zoom).

Column reference — total_boma_count_by_date.csv

Column	Description
`date`	Calendar date on which boma locations were recorded.
`total_count`	Number of mobile boma location events recorded on that date. The final row is a grand Total.

Column reference — mobile_boma_summary_table.csv

Column	Description
`date`	Calendar date of the cattle-count patrol.
`zone_1`	Cattle headcount recorded in Grazing Zone 1 on that date.
`zone_2_3`	Cattle headcount recorded in Grazing Zones 2 & 3 combined on that date.
`zone_4`	Cattle headcount recorded in Grazing Zone 4 on that date.
`total_count`	Sum of cattle across all zones for that date.

Column reference — livestock_predation_events.csv / .gpkg

Column	Description
`id`	Unique EarthRanger event identifier.
`time`	Full timestamp of the predation event.
`date`	Calendar date extracted from the event timestamp.
`event_type`	Event type code (livestock_predation_rep).
`event_category`	Broader ER event category.
`reported_by`	Name of the ranger or user who reported the event.
`geometry`	Point geometry of the predation location.
`suspected_predator`	Predator species suspected to be responsible (e.g. lion, leopard, hyena).
`livestock_species`	Species of livestock killed or attacked (e.g. cattle, sheep, goat).
`no_affected`	Number of animals killed or injured in the incident.

Column reference — livestock_predation_summary_table.csv

Column	Description
`date`	Calendar date of the predation event.
`suspected_predator`	Predator species identified or suspected.
`livestock_species`	Livestock species lost or attacked.

`no_affected`	Total number of animals affected in that event row.

Column reference — livestock_events_recorded_by_date.csv

Column	Description
`date`	Calendar date on which predation events were recorded.
`no_of_events`	Number of livestock predation events recorded on that date. The final row is a grand Total.

Notes

- The predation map uses **fixed coordinates** centered on MNC rather than auto-zooming to the data extent, ensuring consistent map framing across reporting periods.
- If no predation events are recorded for the period, the predation outputs are skipped gracefully and the boma outputs are still produced.
- Zone designations (zone_1, zone_2_3, zone_4) are derived from the MNC grazing-zone boundaries layer; cattle counts are attributed to the zone whose polygon contains each event point.

Section 6: Mara North Event Report

mara_north_event_report

Overview

The Mara North Event Report is the **master combined workflow**. It runs all five component workflows in sequence — TAHMO weather, patrol effort, livestock monitoring, wildlife incidents & sightings, and logistics — producing the full set of CSV, GeoPackage, and map outputs described in Sections 1–5. Once all component outputs have been generated and all HTML maps converted to PNG, the workflow calls the **generate_mnc_report** task to merge every output into a single, formatted Word document using the official MNC report template. This is the recommended entry point when a complete monthly or periodic report is needed.

Required inputs

- EarthRanger credentials and URL (set in workflow parameters).
- Date range: **since** and **until** timestamps.
- MNC conservancy boundaries GeoPackage (auto-downloaded from Dropbox).
- MNC parcels GeoPackage (auto-downloaded from Dropbox).
- **mara_north_event_template.docx** — Word report template (auto-downloaded from Dropbox). Contains pre-formatted section headings, table placeholders, and image frames at 6.9 cm × 11.5 cm.

How it works (step summary)

#	Step	Detail
1	TAHMO weather	Fetches TAHMO station data and produces weather summary CSVs and charts (see Section 1).
2	Patrol effort	Processes patrol trajectories, coverage grid, and event summaries (see Section 2).
3	Livestock monitoring	Processes mobile boma, cattle-count, and livestock predation events (see Section 5).
4	Wildlife incidents	Processes snare, fire, carcass, injury, treatment, and illegal grazing events (see Section 4).
5	Wildlife sightings	Processes elephant, buffalo, rhino, lion, leopard, cheetah, giraffe, and hartebeest sightings (see Section 4).
6	Logistics	Processes balloon landings, airstrip operations, and airstrip maintenance (see Section 3).
7	Convert all HTML to PNG	Converts all HTML maps and charts from every sub-workflow to PNG using Playwright at 2× device scale.

#	Step	Detail
8	Generate Word report	Calls generate_mnc_report with the MNC template to embed all PNG maps and CSV tables into the formatted Word document. Images are sized at 6.9 cm × 11.5 cm within their placeholders.

Output files

All outputs from Sections 1–5 are produced, plus the final consolidated Word document:

Output File	Type	Description
`All Section 1 outputs`	Various	Full TAHMO weather report outputs (CSVs, HTML/PNG charts). See Section 1.
`All Section 2 outputs`	Various	Full patrol effort report outputs (CSVs, GeoParquet, HTML/PNG maps). See Section 2.
`All Section 5 outputs`	Various	Full livestock monitoring outputs (CSVs, GeoPackage, HTML/PNG maps). See Section 5.
`All Section 4 outputs`	Various	Full wildlife monitoring outputs (CSVs, GeoPackages, HTML/PNG maps). See Section 4.
`All Section 3 outputs`	Various	Full logistics report outputs (CSVs, HTML/PNG maps). See Section 3.
`mara_north_event_report_[date].docx`	Word (.docx)	Consolidated MNC event report. All PNG maps and CSV tables embedded in the official template. Filename includes the reporting-period end date.

Word document structure

The **mara_north_event_report_[date].docx** file is assembled from the *mara_north_event_template.docx* template. The template defines the document layout, section headings, and image/table placeholder locations. The **generate_mnc_report** task fills in the placeholders with the outputs from each sub-workflow in the order below:

Report section	Content inserted
Weather summary	TAHMO station rainfall and temperature charts and daily summary table.
Total events	Patrol event counts (excluding distance counts and airstrip events).

Livestock	Mobile boma daily count, cattle-count zone table, boma movement map, predation summary, predation map.
Wildlife incidents	Incident count table, daily incident count, incidents map.
Wildlife sightings	Per-species count tables, herd/individual summaries, and sighting maps (elephant, buffalo, rhino, lion, leopard, cheetah, giraffe, hartebeest).
Logistics	Balloon landing counts, airstrip arrival/departure table, airstrip maintenance summary, logistics map.
Patrol effort	Patrol trajectory map, coverage grid map, patrol info CSV embedded as a table.

Notes

- Running the Mara North Event Report workflow is equivalent to running Sections 1–5 individually plus the Word document generation step.
- Image placeholders in the template accept PNG files at **6.9 cm × 11.5 cm**. Maps are rendered to this size during the HTML-to-PNG conversion step.
- If any task block has no matching events for the reporting period, that section is skipped gracefully and the Word document is still generated — the corresponding section will be blank or contain a 'no data' note.
- The output filename includes the reporting-period end date (derived from the *until* parameter) so consecutive reports do not overwrite each other.